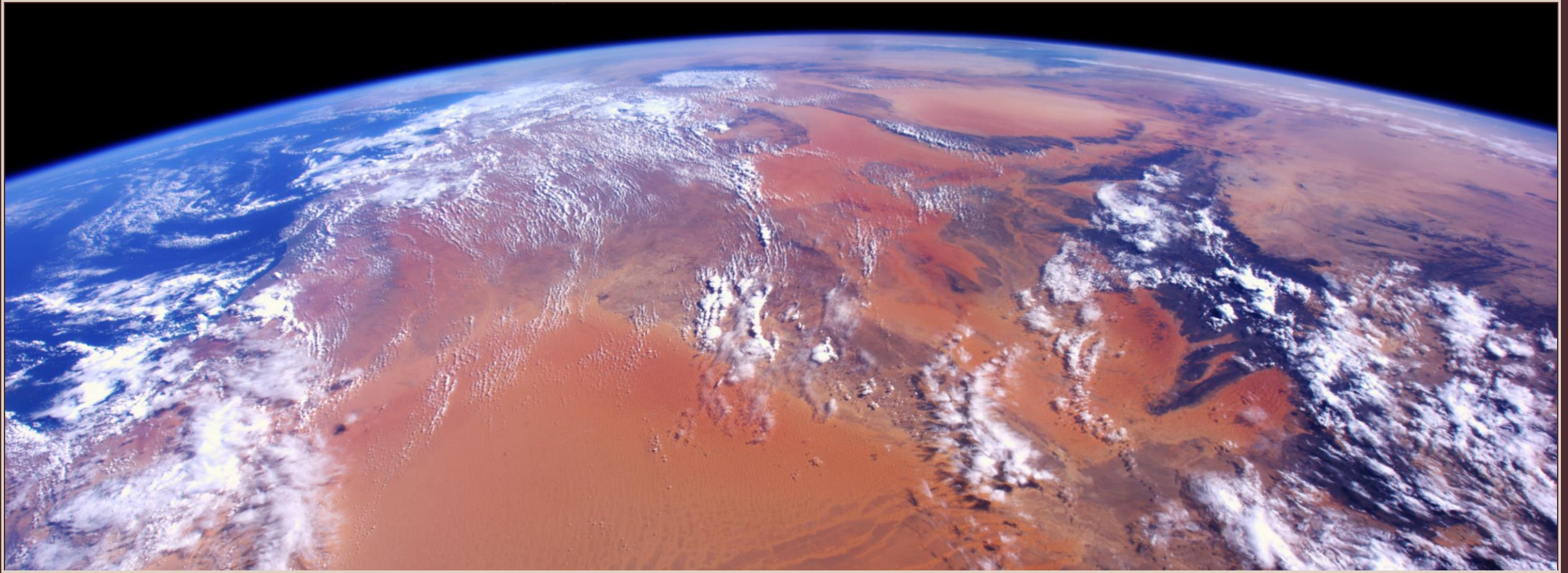




RECAP —
Rapid Event-level Classification of Affected Properties

By: Yatin Raju, Aryan Anand, Sriradhaa Satishkumar

The Problem

Why This Matters

- Natural disasters damage tens of thousands of buildings.
- Manual inspection from satellite imagery is slow and inconsistent.
- Emergency responders need fast, automated, building-level assessments.

Our Goal

- Build an AI system that predicts building damage severity after a disaster using pre- and post-satellite images.

Limitations and Constraints

Model-Related Limitations

- **Underperformance on minority classes**
Minor-damage and major-damage classes have less training data, leading to lower recall and F1 scores.
- **Sensitivity to polygon quality**
Inaccurate or incomplete building polygons in the dataset can reduce prediction reliability.
- **No geographic generalization guarantee**
Model trained on xBD may not fully generalize to regions with different architecture or disaster profiles.

Data & Infrastructure Constraints

- **Large dataset size**
Training requires significant storage and preprocessing time; GitHub cannot host raw data.
- **GPU dependency for speed**
Full test-set evaluation takes several minutes on CPU; real-time use ideally requires GPU acceleration.
- **Label imbalance**
Severe-damage cases are far less common, which biases the model toward “no-damage.”

What Our System Does

High-Level Pipeline

1. **Input:** Pre- & post-disaster satellite tiles
2. **Extract Buildings:** Use segmentation + polygon labels
3. **Compare Pre vs Post:** Track visual changes per building
4. **Predict Damage:**
 - No damage
 - Minor damage
 - Major damage
 - Destroyed
5. **Output:** Color-coded building-level damage maps

Simple, fast, and fully automated.

Pre image



Post image

Dataset Overview (xBD)

xBD Dataset Includes:

- Pre-disaster image
- Post-disaster image
- Building polygon footprints
- 4-class damage labels
- 15 global disasters (fires, hurricanes, floods, earthquakes)

Scale:

- ~45,000 labeled chips
- 850,000+ building polygons

Model Architecture

Siamese ResNet-34 (Pre/Post)

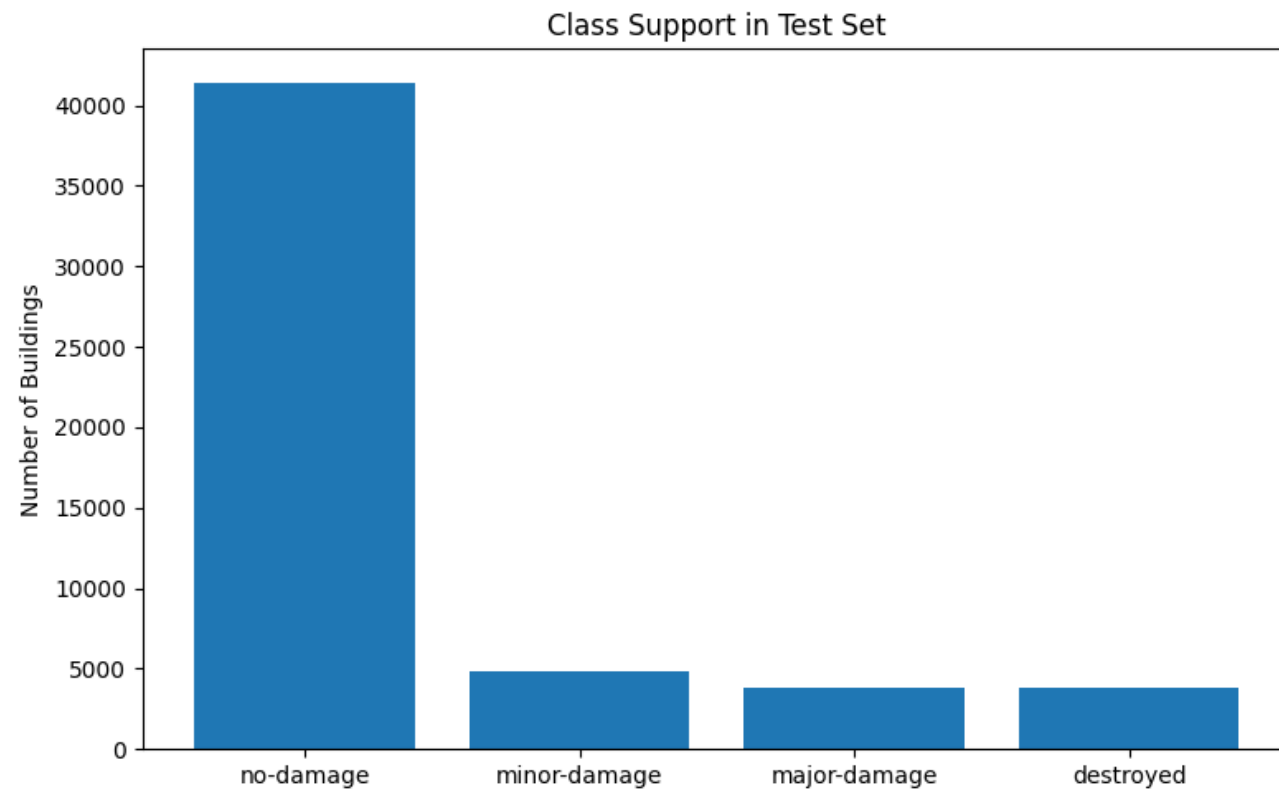
- Two shared ResNet-34 encoders
- Extract features from pre and post images
- Compute difference vectors
 - v_{pre}
 - v_{post}
 - $v_{post} - v_{pre}$
 - $|v_{post} - v_{pre}|$

Polygon-Aware Reasoning

- For each building footprint:
 - Mask $\rightarrow$ pool $\rightarrow$ feature vector

Classifier

- 3-layer MLP
- Outputs one of 4 damage classes



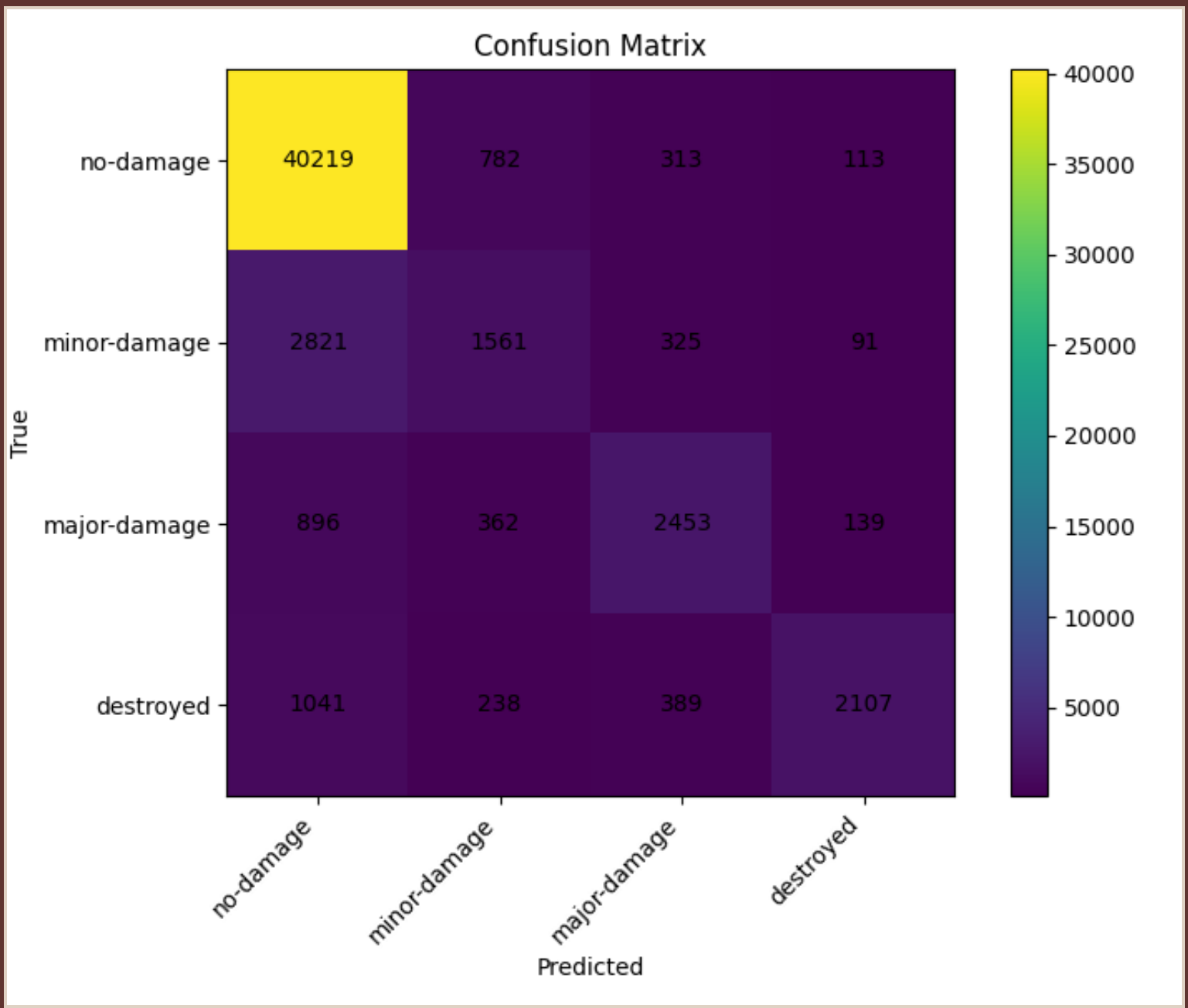
Training Performance Summary

Overall Test Performance

- **Accuracy:** 0.8605
- **Macro F1:** 0.6701
- **Weighted F1:** 0.8474

Class Performance

- No-damage: **F1 = 0.93**
- Minor-damage: **F1 = 0.40**
- Major-damage: **F1 = 0.67**
- Destroyed: **F1 = 0.68**



Confusion Matrix

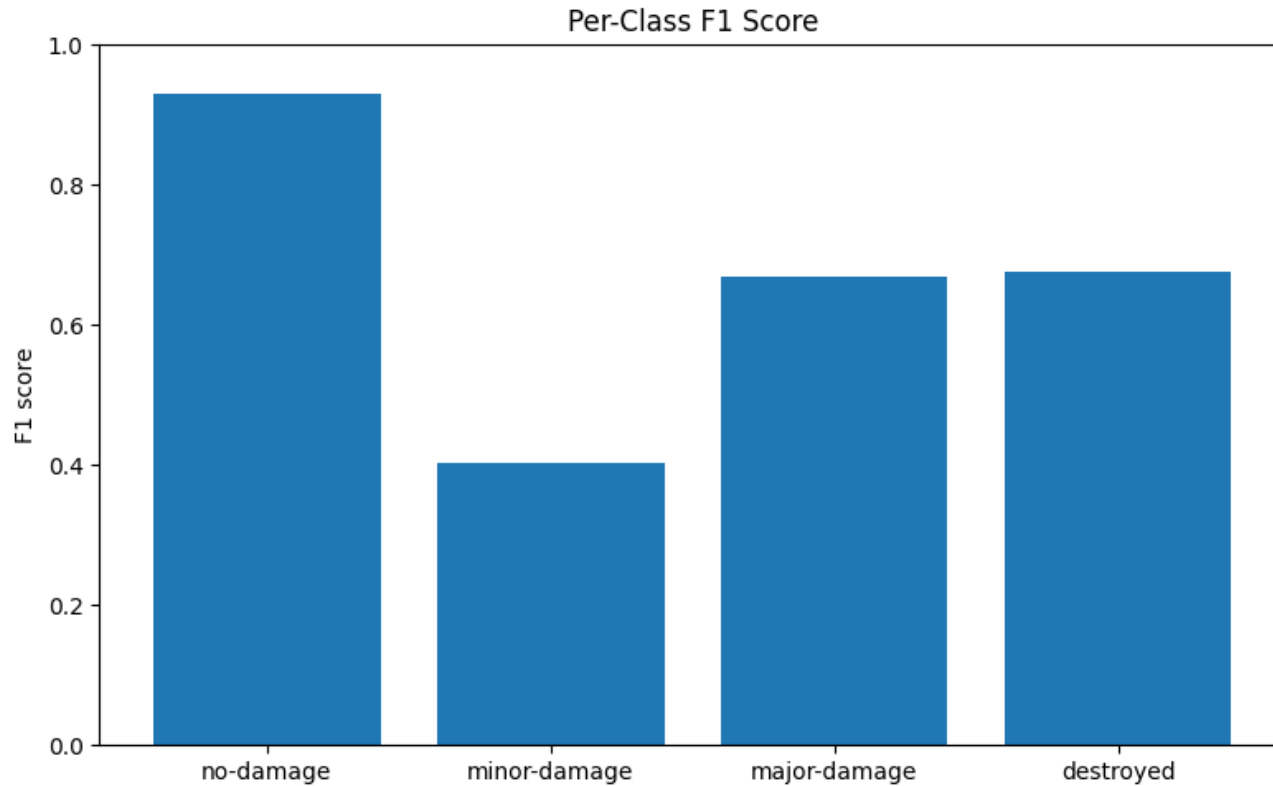
Key Takeaways

- Very strong performance on "no-damage"
- Good performance on "destroyed"
- Confusion mainly between *minor* and *major* damage
- Minor damage is the hardest class (expected due to few examples)

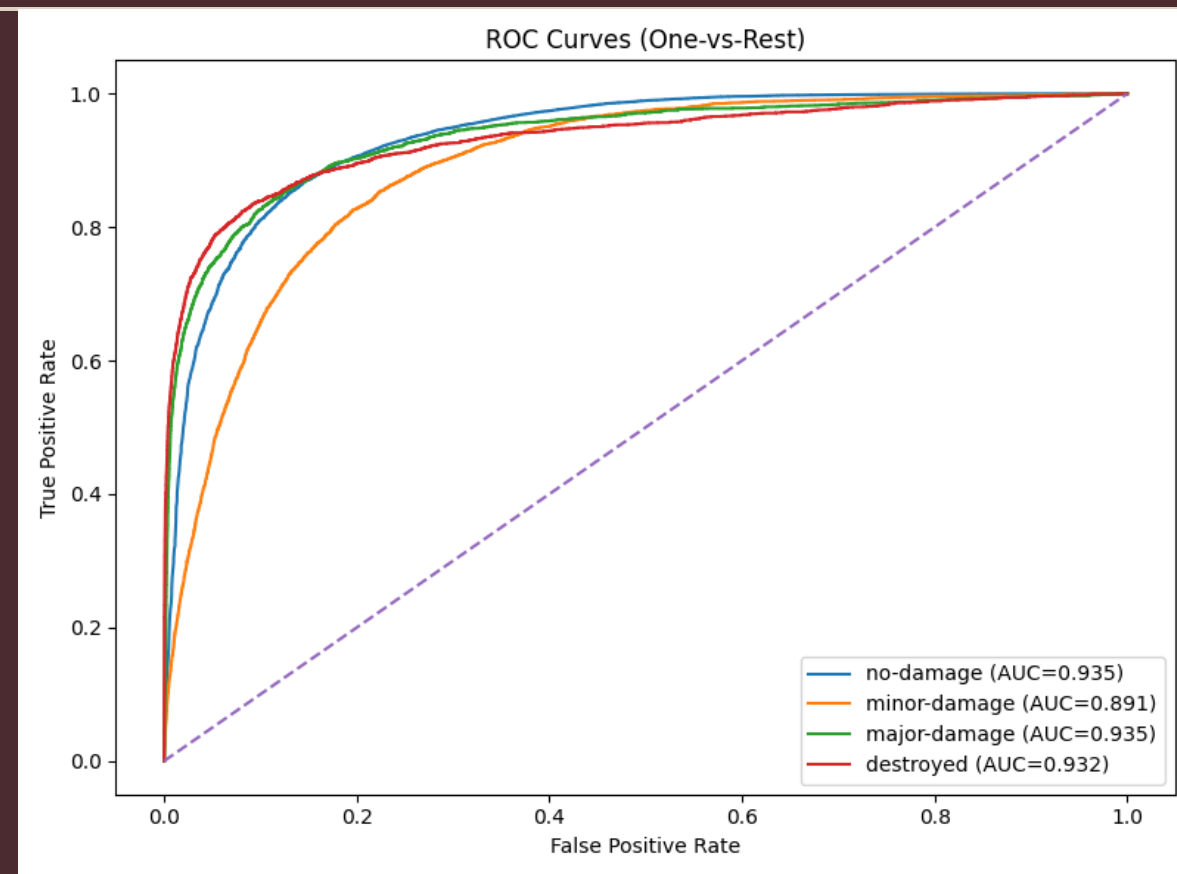
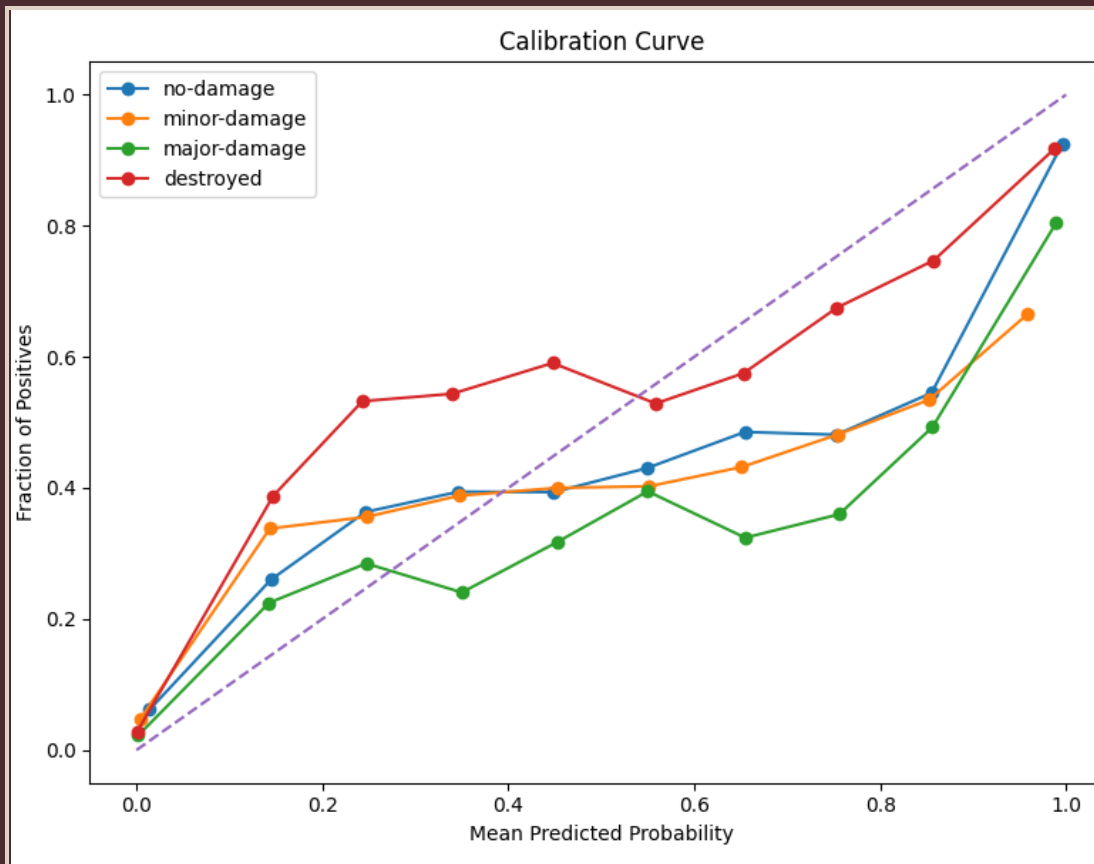
Per-Class Model Quality

Observations

1. High F1 for no-damage
2. Strong for destroyed
3. Medium for major damage
4. Weak for minor damage
(dataset imbalance + subtle visual differences)



Model Reliability & Confidence



Insights

- Good probability reliability (curve close to diagonal)
- High AUC for no-damage & destroyed
- Lower AUC for minor-damage → reflects difficulty of subtle changes
- Model confidence generally aligns well with correctness

Frontend Demo

[Video Link](#)

Thanks for
watching

